THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. If A : B = 2 : 3 and B : C = 6 : 5, then A : B : C is: 1

(a) 4 : 6 : 5 **(b)** 2 : 3 : 5 **(c)** 2 : 6 : 5 **(d)** 2 : 3 : 30

2. 18 is what percent of 72? 1

(a) 25% **(b)** 22% **(c)** 28% **(d)** 50%

3. The legs of a right triangle are 7 cm and 9 cm. The area of the square drawn on its hypotenuse (in cm²) is: 1

(a) 130 **(b)** 16 **(c)** 256 **(d)** 63

4. 18 men can do a piece of work in 10 days. 12 men will do it in (days): 1

(a) 15 **(b)** 11 **(c)** 18 **(d)** 180

5. A Koch curve starts with one segment. The number of segments at stage 1 is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 3

6. The mean of 5 numbers is 16. If 4 of them are 12, 18, 20, 10, the remaining number is: 1

(a) 25 **(b)** 16 **(c)** 15 **(d)** 20

7. The sum of 3 consecutive integers starting with n is: 1

(a) $3n + 4$ **(b)** $3n + 3$ **(c)** $n + 3$ **(d)** $3n$

8. The area of a triangle is 72 cm² and its base is 16 cm. Its height (in cm) is: 1

(a) 18 **(b)** 11 **(c)** 12 **(d)** 9

9. The triangular number T_8 (sum of 1 to 8) is: 1

(a) 64 **(b)** 36 **(c)** 28 **(d)** 44

10. The standard form of 60000000 is: 1

(a) 6×10^7 **(b)** 6×10^6 **(c)** 6×10^8 **(d)** 60×10^7

11. The number 87 written in base 5 is: 1

(a) 223 **(b)** 321 **(c)** 323 **(d)** 322

12. In rhombus ABCD, $\angle ABC = 70°$. The angle $\angle OBC$ (O is the point where the diagonals meet) is: 1

(a) $70°$ **(b)** $35°$ **(c)** $110°$ **(d)** $55°$

13. For the number 2538, (largest 4-digit arrangement of its digits) - (smallest arrangement) is: 1

(a) 6084 **(b)** 6264 **(c)** 6174 **(d)** 6183

14. $(x + 3)(x - 3)$ equals: 1

(a) $x^2 + 9$ **(b)** $x^2 - 6x + 9$ **(c)** $x^2 + 6x - 9$ **(d)** $x^2 - 9$

15. On a map, 1 cm represents 5 km. A distance of 3.5 cm on the map is actually: 1

(a) 8.5 km **(b)** 22.5 km **(c)** 17.5 km **(d)** 1.75 km

16. A cycle bought for ₹ 1500 is sold for ₹ 1800. The gain percent is: 1

(a) 22% **(b)** 20% **(c)** 40% **(d)** 18%

17. Taking $m = 8$ in $(2m,\ m^2 - 1,\ m^2 + 1)$, the Pythagorean triplet is: 1

(a) 16, 64, 65 **(b)** 8, 63, 65 **(c)** 16, 63, 65 **(d)** 16, 63, 66

18. On a map the scale is 1 cm = 15 km. Two places 135 km apart are shown on the map at a distance (in cm) of: 1

(a) 9 **(b)** 10 **(c)** 2025 **(d)** 120

19. In the Sierpinski triangle (area of the original triangle taken as 1), the fraction of area still shaded after stage 2 is: 1

(a) $\frac{9}{16}$ **(b)** $\frac{8}{16}$ **(c)** $\frac{1}{16}$ **(d)** $\frac{6}{8}$

20. The mean of 5 observations 39, 11, 19, 36, 15 is 24. If a new observation 12 is added, the new mean is: 1

(a) 36 **(b)** 27 **(c)** 22 **(d)** 24

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Find the ratio of 3 m 60 cm to 90 cm in simplest form. 2

22. What percent of 90 is 36? 2

23. Find the Pythagorean triplet with the smallest member 28. 2

24. Which ratio is greater: 5:9 or 3:5? 2

25. A prism has 21 edges. Find the number of sides of its base and the number of faces. 2

26. Find the median of 21, 12, 34, 44, 45, 38. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. The hypotenuse of an isosceles right triangle is $4\sqrt{2}$ cm. Find the length of each equal side and the area of the triangle. 3

28. On a map of scale 1 cm = 2 km, a rectangular park measures 7 cm by 4 cm. Find the actual length, breadth and perimeter of the park. 3

29. A cube of side 3 cm is painted on all faces and cut into 1 cm cubes. Find the number of small cubes with 3

(i) exactly two faces painted

(ii) exactly one face painted.

UNDER MIND

30. The runs scored by a batter in 6 innings are 78, 80, 89, 55, 31, 33. Find the mean, the median and the range. 3

31. In a number pyramid with bottom row 2, 3, 1, 4, each block is the sum of the two below it. Find all the blocks. 3

OR

A rectangular lawn is 32 m by 18 m. Find its area, and the cost of levelling it at ₹ 12 per m². 3

32. Check whether $(22,\ 120,\ 122)$ is a Pythagorean triplet. 3

OR

Find the value of $\left(\frac{2}{3}\right)^{-1} \times \left(\frac{2}{3}\right)^{-1}$. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. ₹ 5250 is divided among A, B and C such that A:B = 2:3 and B:C = 4:5. Find the share of each. 5

34. Find the number of faces, vertices and edges of 5
(i) a triangular prism and
(ii) a triangular pyramid, and verify Euler's formula for both.

35. The monthly rainfall (mm) from June to November is 181, 83, 44, 173, 148, 151. 5
(i) Find the mean rainfall.
(ii) Which month had the greatest rise over the previous month?
(iii) Find the range.

36. Show that the sum of 3 consecutive integers starting with n is $3n + 3$ and verify it for $n = 14$. 5

OR

A trapezium has parallel sides 20 cm and 12 cm and the distance between them is 9 cm. Find its area. A parallelogram of the same height and base 12 cm (the shorter parallel side) is cut from it, leaving a triangle: find the area left over. 5

37. Find $\sqrt[3]{13824}$ by prime factorisation and verify the result by cubing. 5

OR

Write 600000 and 2000 in standard form. Then find their product and quotient in standard form. 5

38. A shopkeeper has weights of 1, 2, 4, 8, 16 and 32 kg. Show how to weigh exactly 13 kg using these weights, each at most once. Explain the idea using binary numbers. 5

OR

(i) State two properties of the diagonals of a rhombus. 5
(ii) In rhombus ABCD the diagonals meet at O and $\angle BAD = 56°$. Find $\angle OAB$, $\angle OBA$ and $\angle ABC$.

END OF QUESTION PAPER — Total: 80 Marks